Daksh Chhillar

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B.Tech
Chemical
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (Chemical)	Indian Institute of Technology, Delhi	7.56(Current)	2024-Present
Senior Secondary	CBSE Board	95.0%	2023
Secondary	CBSE Board	96.5%	2021

Scholastic Achievements

- **Joint Entrance Examination 2024**
  - **JEE Advanced:** Secured **AIR 1635 (CRL)** out of 180,200 candidates with a score of **215/360** marks.
  - **JEE Mains:** Secured **AIR 8667 (CRL)** among more than 10 Lakh candidates with **99.46 percentile**.
- Science Olympiad Foundation(SOF): Won multiple medals in IMO, ISO; stood among the **top 1%**

Projects

- **Adaptive Step-Size Methods for Stiff ODE Systems** *Nov. 2025*
  Prof. Jayati Sarkar
  - Designed a hybrid adaptive solver improving stability and efficiency for stiff chemical kinetics simulations.
  - Implemented and compared Implicit Euler, Rosenbrock, and Dormand–Prince RK45; developed a hybrid RK45–Rosenbrock solver with regression-based step prediction for a non-isothermal CSTR ignition model framework.
  - Analyzed instability of explicit ODE solvers in stiff chemical kinetics systems with rapid time-scale variations.
  - Exposure to stiff ODE solvers, adaptive time-stepping algorithms, reactor modeling, and numerical simulation.
- **Viscous Dissipation Effects in Laminar Channel Flow** *Nov. 2025*
  Prof. Vikrant Verma
  - Analyzed the impact of viscous dissipation on temperature profiles in fully developed laminar channel flow.
  - Derived the energy equation including viscous heating and performed Brinkman and Péclet number analysis.
  - Computed temperature profiles for isothermal and adiabatic walls using analytical and finite-difference approaches.
  - Developed exposure to heat transfer modeling, transport phenomena, numerical methods, and fluid mechanics.
- **3D Turbulent Flow Simulation in an Ellipsoidal Micro-Conduit** *Ongoing*
  Prof. Somnath Ghosh
  - Simulated turbulent water flow through an ellipsoidal micro-conduit and analyzed velocity fields and pressure losses.
  - Computed normalized velocity profiles and compared them with theoretical predictions to assess flow behavior.
  - Evaluated pressure drop variation with maximum velocity under steady-state flow conditions in the conduit.
  - Gained exposure to CFD simulations, mesh independence testing, numerical fluid mechanics, and flow visualization.

Key Courses Taken

- Numerical Methods, Transport Phenomena, Material & Energy Balances, Linear Algebra, Calculus, Introduction to Computer Science, Chemistry of Interfaces, Chemical Engineering Thermodynamics, Chemical Reaction Engineering, Fluid Mechanics, Heat Transfer

Technical Skills

- **Programming and Web Skills:** C/C++, Python, SQL, HTML/CSS
- **Tools and Frameworks:** Simulation(COMSOL Multiphysics, ANSYS Fluent), Figma, Canva, FreeCAD, NumPY

Positions of Responsibility and Co-Curricular Activities

- **Secretary, Photography and Films Club,** Board of Cultural and Recreational Activities(BRCA),IIT Delhi, Present
- **Hostel Representative, Photography and Films Club,** BRCA, IIT Delhi, 2025-26
- **Coordinator,** Entrepreneurship Development Cell's annual fest, BeCON 2026
- **Executive,** Chemical Society(ChES), IIT Delhi, Present
- **Production Head,** IITD OnAIR, Present
- **Secured 2st Position** General Championship 2024-2025, BRCA, IITD
- Represented IIT Delhi in **Inter IIT Cultural Meet 2025** in filmmaking. Secured **2nd position** competing with **20+ IITs**
- **Secured 2st Position** General Championship, BRCA, IITD
- **Winner** Design Club Trophy and Dance Club Trophy as part of Niligiri hostel team,2024-2025, BRCA, IITD
- **Podium finish** for producing an original short film for Board for Student Publications, IITD, 2025